

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

8th Semester of B. Pharm. Examination

University Theory Examination April/May 2016

PH410 Medical Chemistry – IV

Date: 09.05.2016, Monday

Time: 10:00 a.m to 01:00 p.m

Maximum Marks: 80

Instructions:

1. Figures to right indicate marks.
2. Section wise separate answer book to be used.
3. Draw neat sketches wherever necessary

SECTION- I

Q 1 Attempt any **TWO** of the following

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|---|-----|--|----|
| A | I. | What is β -lactams? Explain SAR of Penicillin derivatives in details. | 05 |
| | II. | Write a chemistry of Aminoglycosides in details. | 05 |
| B | | Enlist various approaches of combinatorial library synthesis. Explain parallel synthesis in details. Give applications of combinatorial chemistry. | 10 |
| C | I. | What is combinatorial Chemistry? Explain Principle involved in Combinatorial chemistry | 05 |
| | II. | Classify polyene & polypeptide antibiotics with their structure. | 05 |

Q 2 Attempt any **TWO** of the following

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|---|-----|--|----|
| A | I. | Write a detail note on Quantitative Structure Pharmacokinetic Relationship (QSPR). | 05 |
| | II. | Write an application of bioinformatics. | 05 |
| B | I. | Write a mechanism of action and SAR of Chloramphenicol in details | 05 |
| | II. | Explain SAR of Cephalosporins. | 05 |
| C | I. | Give SAR of tetracyclines. | 05 |
| | II. | Write a short note on macrolide antibiotics. | 05 |

SECTION- II

Q 3 Attempt any **TWO** of the following

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|---|-----|---|----|
| A | I. | What are antifungal agents? How they act? Give synthesis of Fluconazole. | 05 |
| | II. | Write a note on anti-HIV agents. | 05 |
| B | I. | Classify antimalarials with examples. Explain SAR of Cinchona Alkaloids as antimalarial agents. | 05 |
| | II. | Explain SAR of Sulphonamides in details. | 05 |
| C | I. | What are Quinolones? Write structure activity relationship of Quinolones. | 05 |
| | II. | Write a short note on Immunosuppressive agent | 05 |

Q 4 Attempt any **TWO** of the following

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|---|-----|---|----|
| A | I. | What is amoebiasis? How antiamoebic act? Give synthesis of Metronidazole. | 05 |
| | II. | Write short note on antiseptics. | 05 |
| B | | What are antineoplastic agents? Classify in detail with example. Give synthesis of methotrexate and 6-mercaptopurine. | 10 |
| C | I. | Give synthesis of the following compounds : (i) Isoniazid (ii) Ethambutol | 05 |
| | II. | What are anti-TB agents? Give first line and second line drugs used in TB | 05 |

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